Design and CFD Simulation of a 600 N Rocket Nozzle

High-Altitude Performance & Compressible Flow Analysis

Software used : Ansys Workbench

By:

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Part I:
Design Requirements & Calculation

Design Parameters

Target Thrust

$F = 600\ N$

Designed for high-thrust short-duration missions.

Chamber Pressure

$P0 = 1.8\ MPa$

Calculated for stable combustion chamber operation.

Inlet Temperature

$T0 = 3000\ K$

Based on standard propellant combustion adiabatic flame temp.

Gas Properties

$\gamma = 1.3$ | $R = 300\ J/kgK$

Typical properties for rocket exhaust products.

Governing Thrust Equations

Thrust in Vacuum Conditions

$$F = \dot{m}V_e + (P_e - P_a)A_e$$

Assuming near-vacuum ambient pressure (Pa ≈ 0), the momentum thrust component dominates the engine's performance.

2900
m/s (Ve)

0.27
kg/s (Mass Flow)

NOZZLE GEOMETRY

3.1 Thrust Equation

$$F = \dot{m}V_e + (P_e - P_a)A_e$$

For vacuum-like condition:

$$F \approx \dot{m}V_e$$

3.2 Exit Velocity

Exit velocity equation:

$$V_e = \sqrt{\frac{2\gamma}{\gamma - 1}RT_0\left[1 - \left(\frac{P_e}{P_0}\right)^{\frac{\gamma-1}{\gamma}}\right]}$$

Assume expansion to near vacuum:

$$P_e/P_0 \approx 0.05$$

Substitute:

$$V_e \approx 2200\, m/s$$

3.3 Mass Flow Rate Required

From thrust equation:

$$\dot{m} = \frac{F}{V_e}$$

$$\dot{m} = \frac{600}{2200}$$

$$\dot{m} = 0.27\, kg/s$$

3.4 Throat Area Calculation

Mass flow rate equation for choked flow:

$$\dot{m} = A_t P_0 \sqrt{\frac{\gamma}{RT_0}}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

Solving for throat area:

$$A_t = \frac{\dot{m}}{P_0}\sqrt{\frac{RT_0}{\gamma}}\left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

Result:

$$A_t \approx 2.0 \times 10^{-4}\, m^2$$

3.5 Throat Diameter

$$A = \frac{\pi d^2}{4}$$

$$d_t = \sqrt{\frac{4A}{\pi}}$$

$$d_t = 16\,mm$$

3.6 Exit Area Calculation

Using area-Mach relation:

$$\frac{A_e}{A_t} \approx 5.23$$

$$A_e = 5.23 \times A_t$$

$$A_e = 1.05 \times 10^{-3} m^2$$

3.7 Exit Diameter

$$d_e = 36.6\,mm$$

3.9 Length Selection

Typical rocket proportions:

Chamber length:

$$8 \text{ to } 12 \times throatdiameter$$

Selected:

$$160mm$$

Convergent length:

$$1.5 \text{ to } 2throatdiameters$$

Selected:

$$28mm$$

Divergent length based on expansion angle:

$$15^\circ$$

Selected:

$$38mm$$

Total length:

$$226mm$$

Final Design Specification

Parameter	Value
Chamber diameter	48 mm
Chamber length	160 mm
Convergent length	28 mm
Throat diameter	16 mm
Divergent length	38 mm
Exit diameter	36.6 mm
Total length	226 mm

Numerical Simulation Strategy

Solver

Pressure-Based Steady Solver

Turbulence

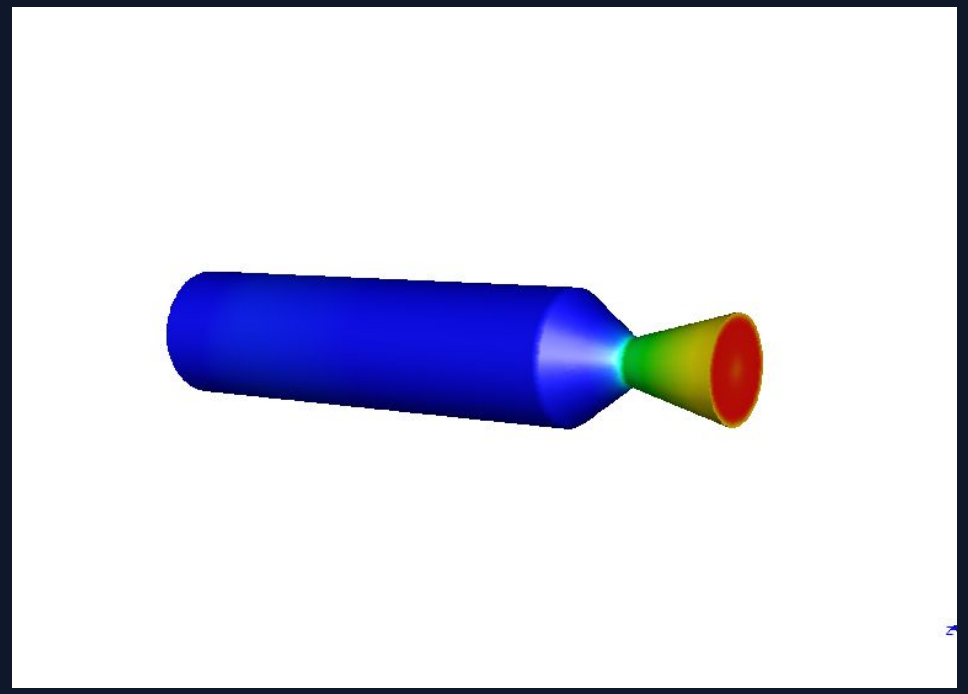
k-omega SST Model

Material

Ideal Gas (Compressible)

Viscosity

Sutherland's Law



Structured High-Resolution Mesh for Nozzle Interior

Governing CFD Equations

Conservation Laws

The solver iteratively solves the Navier-Stokes equations for high-speed compressible gas flow:

- **Continuity:** Mass balance
- **Momentum:** Pressure vs Viscous forces
- **Energy:** Heat-to-kinetic conversion

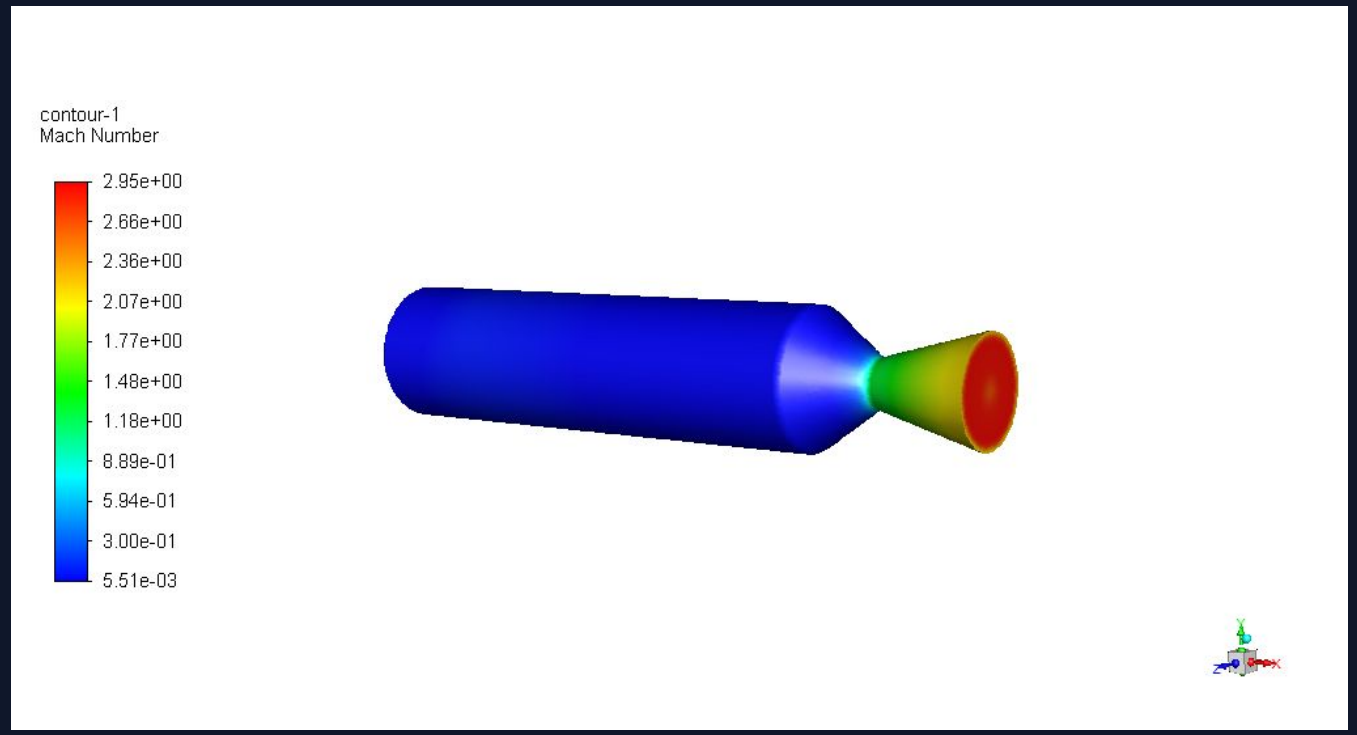
Energy Balance (MathML):

$$\nabla \cdot (\rho V h) = \nabla \cdot (k \nabla T)$$

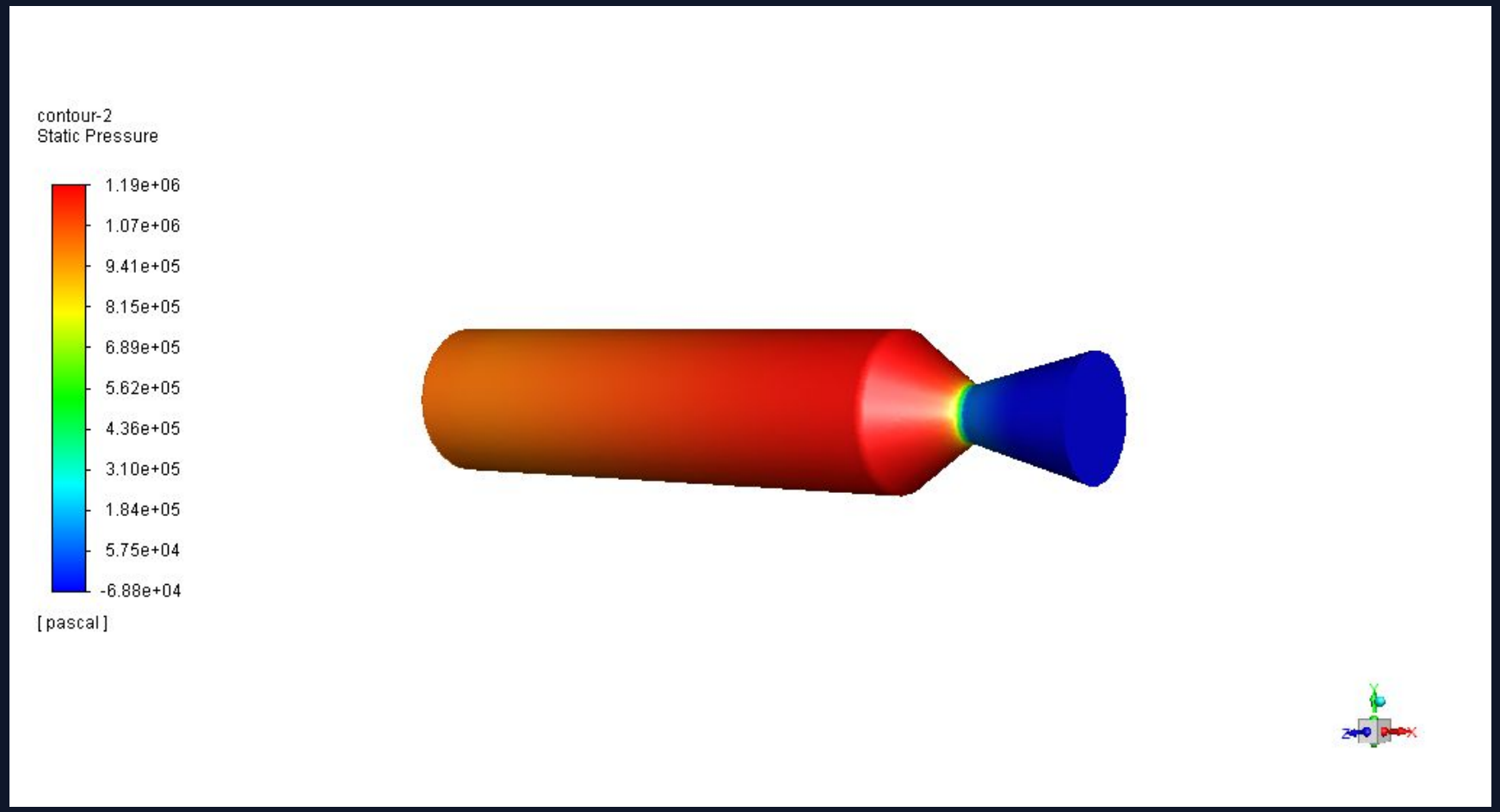
Mach Number Distribution

Supersonic Expansion

The CFD results show a clear transition from subsonic (Mach 0.05) in the chamber to sonic (Mach 1.0) at the throat.
In the divergent section, the flow rapidly accelerates, reaching a peak exit velocity of **Mach 2.9**, confirming successful nozzle design for 600 N thrust.

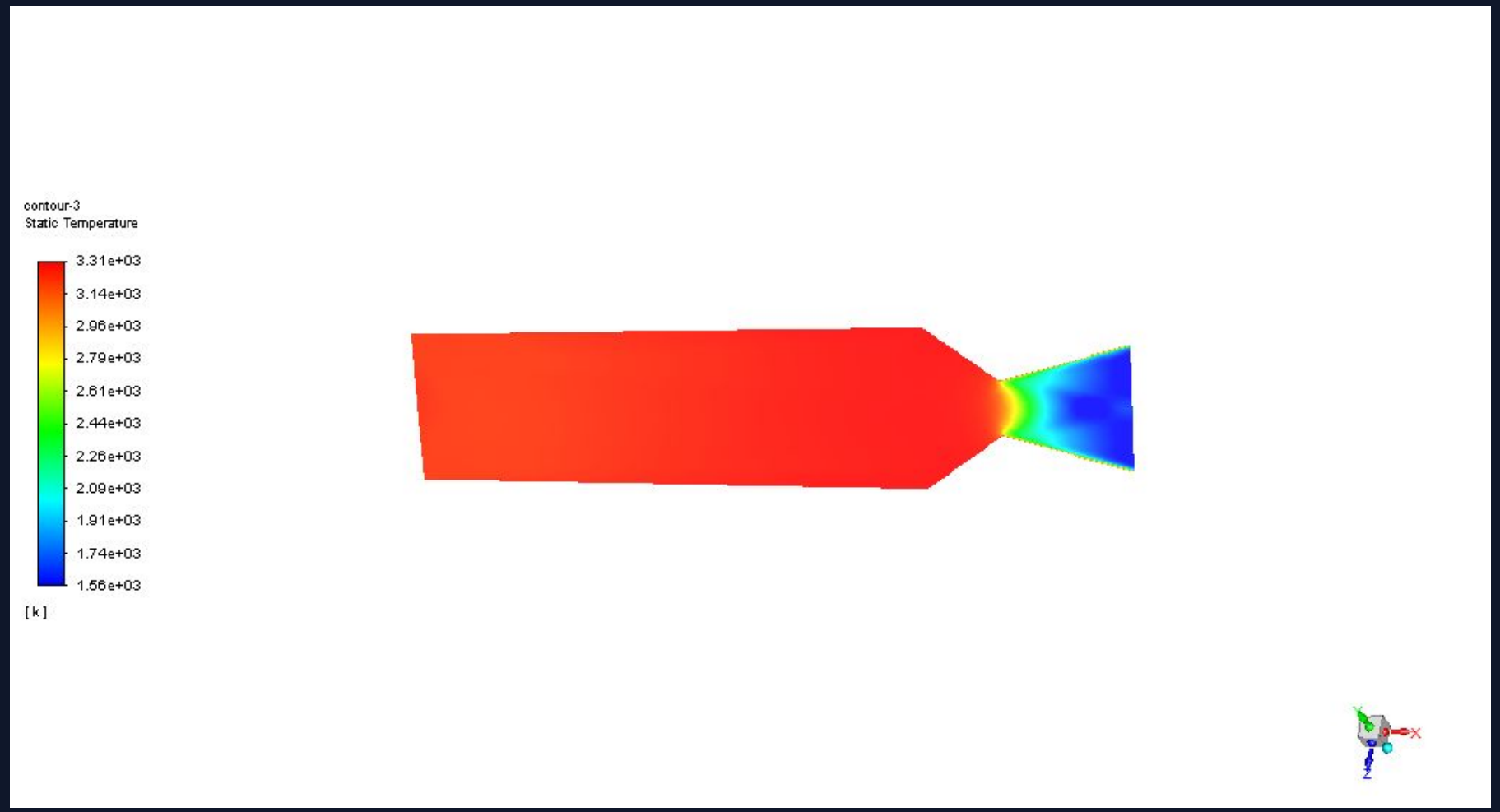


Pressure Validation

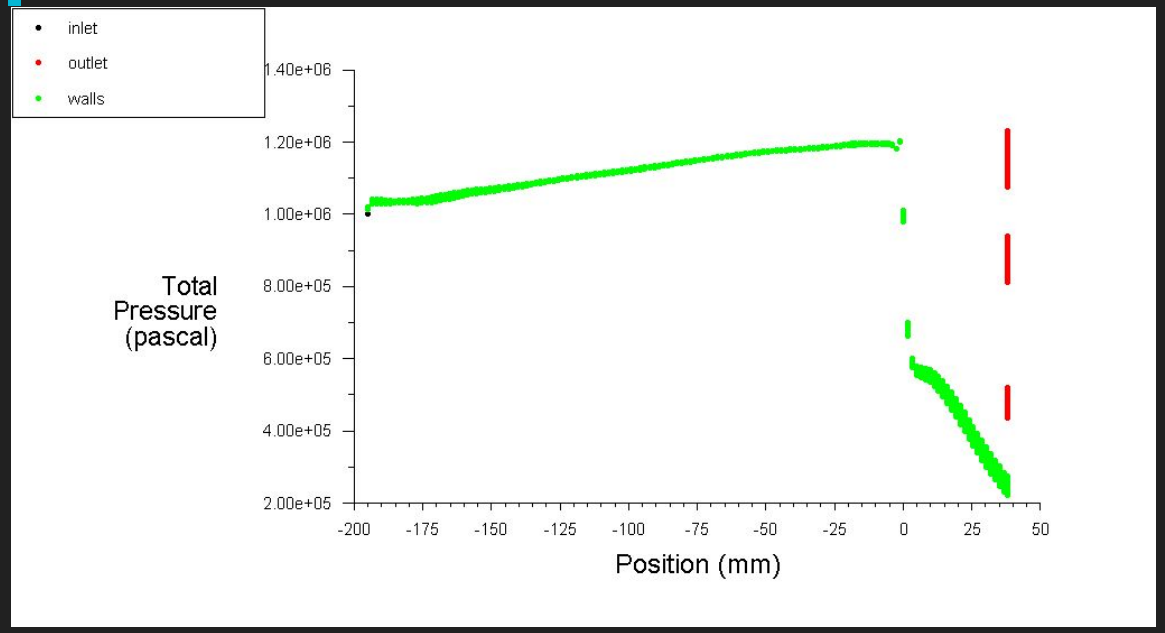


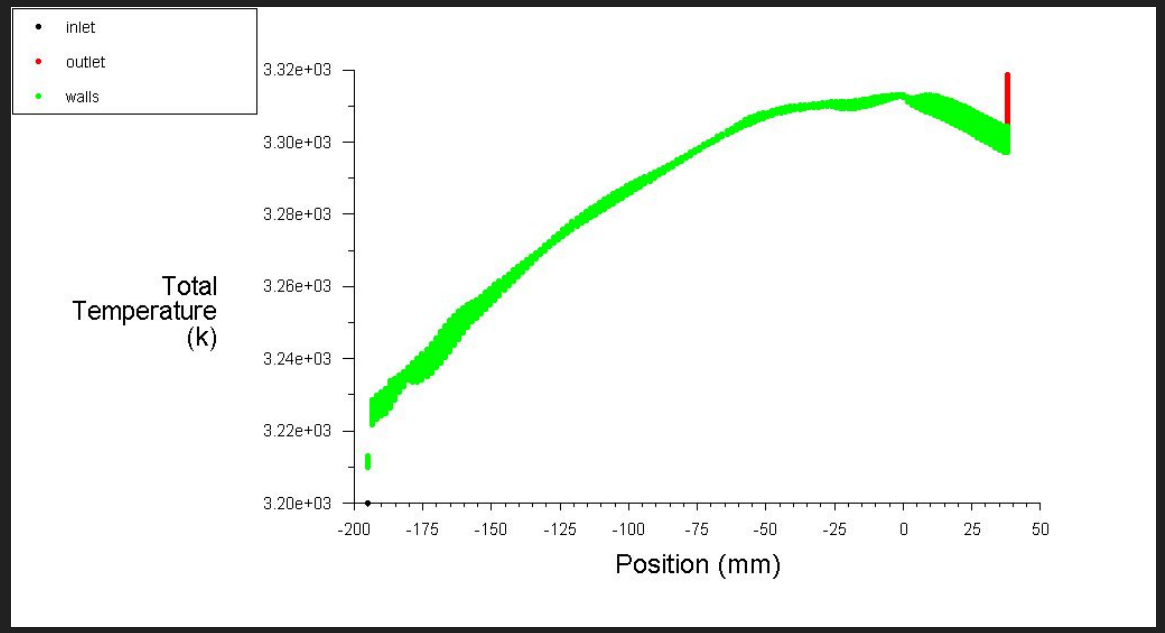
Pressure Contours showing rapid expansion

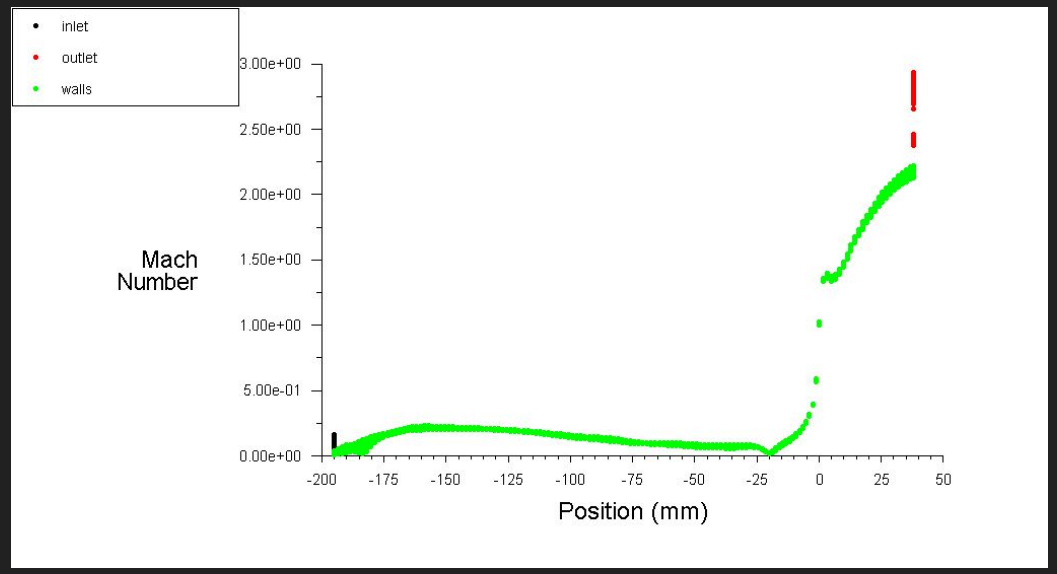
Temperature Validation



Plots Validation







Forces - Direction Vector (1 0 0)

	Forces (n)		
Zone	Pressure	Viscous	Total
walls	734.10433	1.5939795	735.69831
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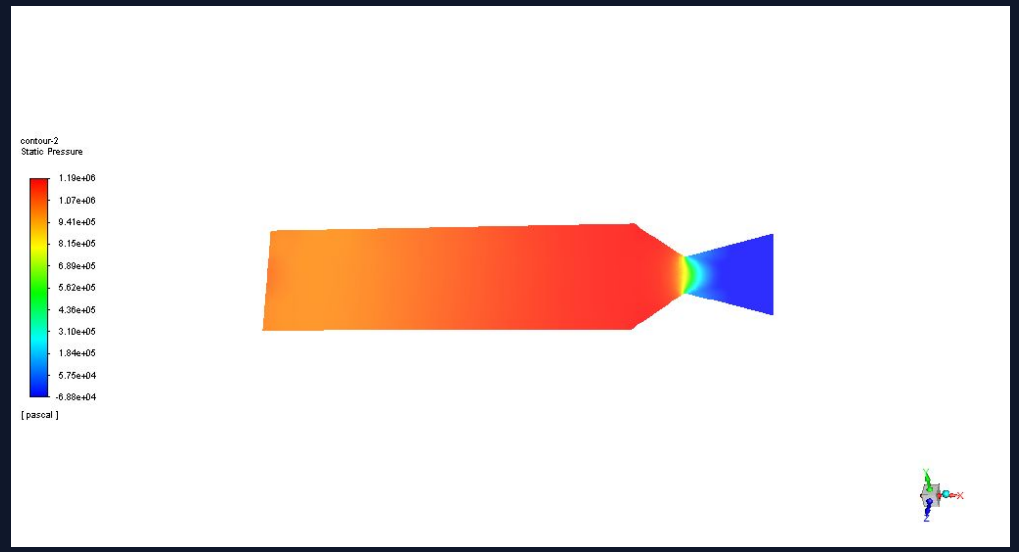
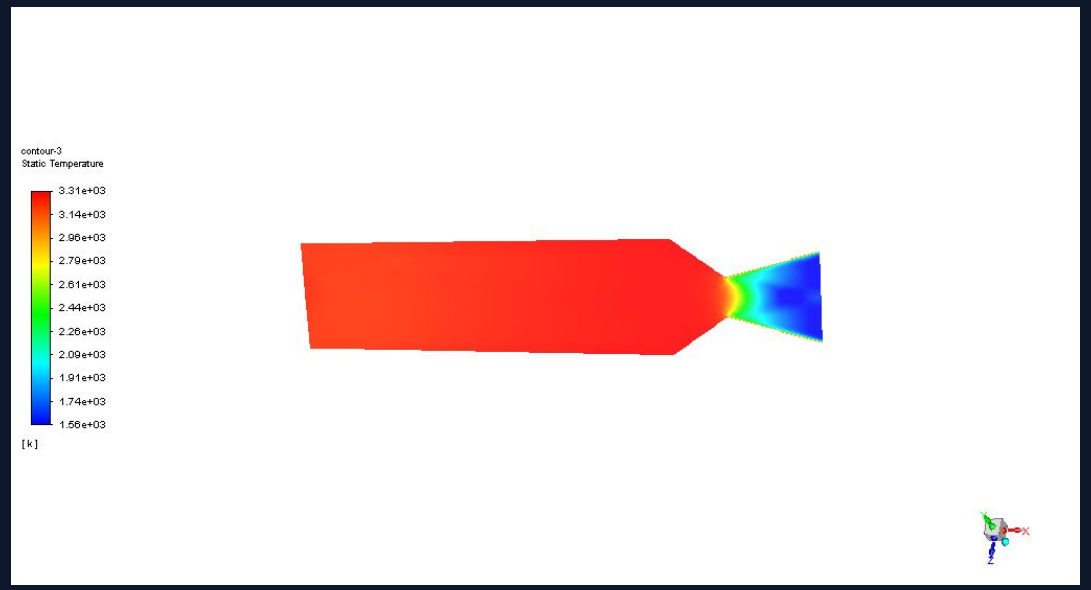
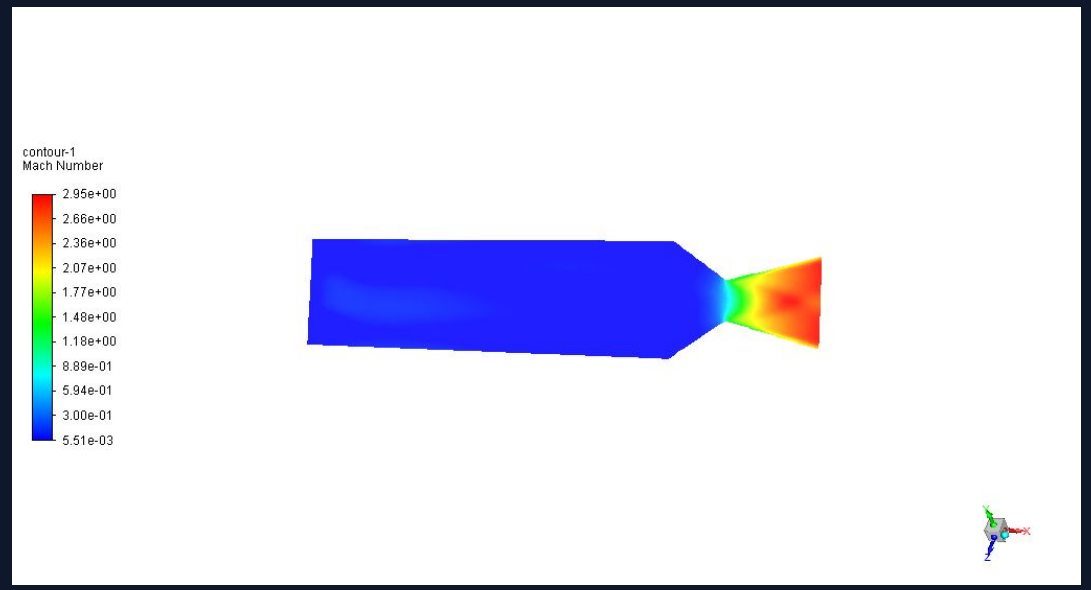
Net	734.10433	1.5939795	735.69831







Conclusion

A convergent–divergent rocket nozzle capable of producing approximately 600 N thrust was successfully designed using compressible flow theory and validated using ANSYS Fluent CFD simulation.

The nozzle geometry was calculated from first principles using thrust, mass flow rate, and area–Mach number relations. The final design achieved an area ratio of 5.23, suitable for supersonic expansion.

CFD results confirmed the expected physical behavior:

- *Subsonic flow in the combustion chamber*
- *Choked flow (Mach 1) at the throat*
- *Supersonic flow (Mach ≈ 2.5–3) at the exit*
- *Significant pressure and temperature drop along the nozzle*
- *Efficient conversion of thermal energy into kinetic energy*

The simulation validated that the nozzle design is capable of generating thrust in the 600 N propulsion class.

Through this project, practical experience was gained in:

- *Rocket nozzle design using compressible flow equations*
- *Geometry creation and meshing for internal flow*
- *Setting up compressible CFD simulations in ANSYS Fluent*
- *Using the k–ω SST turbulence model for high-speed flow*
- *Applying ideal gas and high-temperature material models*
- *Interpreting Mach number, pressure, and temperature contours*

This project strengthened understanding of CFD, turbulence modeling, and aerospace propulsion flow physics, and established a strong foundation for advanced CFD applications in rocket engines, aerospace, and high-speed aerodynamics.